



AMTEC
AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2024 – 0382



Agricultural Machinery – Four-wheel Tractor YTO MF554-6 TIER 3

Test Report Number	2024 – 0382
Date of Issuance	October 7, 2024
Valid Until	October 7, 2029
Agricultural Machinery	Four-wheel Tractor
Type	Four-wheel drive, Four-cylinder, Naturally-aspirated diesel engine, 8F + 8R transmission system, Full hydrostatic steering system
Brand	YTO
Model	MF554-6 TIER 3
Test Requested By	FORD TRACTOR PHILS., INC. 1384 Jose Abad Santos St., Tondo, Manila

SUMMARY

Performance Criteria	Requirement as per PNS/BAFS PABES 301:2020	AMTEC Test
Rated Maximum Power, kW	40.5 ^a	40.5 ^a
Rated Engine Shaft Speed, rpm	2200 ^a	2200 ^a
Maximum PTO Power, kW	38.5 ^a	42.49
Percentage PTO Power of the Engine Power, %, minimum	85	104.92
Specific Fuel Consumption, g/kW-h, maximum	350	236.83 ^b
At Maximum PTO Power		
Engine Speed, rpm	ND ^a	2004
PTO Shaft Speed, rpm	ND ^a	560
PTO Torque, N-m	ND ^a	725.00
Fuel Consumption, L/h	ND ^a	11.98
At 540 rpm PTO Shaft Speed		
Engine Speed, rpm	ND ^a	1860
PTO Torque, N-m	ND ^a	740.67
PTO Power, kW	ND ^a	42.07
Fuel Consumption, L/h	ND ^a	11.75
Specific Fuel Consumption, g/kW-h	ND ^a	234.63

^a The data were obtained from the tractor's nameplate and the specifications sheet provided by the test applicant.

^b Specific fuel consumption at maximum PTO power.

Note: ND – No Data

SUMMARY (Cont'n)

Performance Criteria	Requirement as per PNS/BAFS PABES 301:2020	AMTEC Test
Main Gear	ND ^a	1, 2, 3, 4, N
Range Gear	ND ^a	Low, High
Shuttle Gear	ND ^a	Forward, Reverse
Slowest Forward Speed, kph	ND ^a	2.69
Fastest Forward Speed, kph	ND ^a	29.62
Radius of Turning Circle, mm (Without Brakes)		
Right Turn	ND ^a	4273
Left Turn	ND ^a	4183
Position and/or Draft Implement Controls	Shall be provided	Provided
Three-Point Linkage		
Category based on PTO power at rated rotational frequency of the engine	1: Up to 48 kW 2N/2: 30 – 92 kW	1, 2N/2
Category based on dimensions of tractor linkage points	2N/2: d ₁ : 25.7+0.2 d ₂ : 28.7+0.3	2N, 2
Power-Take-Off Shaft	540 rpm shall have 6 straight splines or 100 rpm shall have 21 involute splines	540 rpm (6 straight splines)
Drawbar Category based on PTO power	1: PTO power ≤48 2: PTO power ≤115 3: PTO power ≤185 4: PTO power ≤200 5: PTO power ≤500	Category 1, 2, 3, 4, and 5
Drawbar Power, kW	ND ^a	36.54 ^b
Hydraulic Lift Force Capacity, kN	11.32 ^c	ND ^d
Wheel Tread Adjustment	Shall be provided	Provided
PTO Casing/Cover	Shall be provided	Provided
Roll-Over-Protective Structure (ROPS)	Shall be provided	Provided
Protective Seat Belt	Shall be provided	Provided

^a The data were obtained from the specifications sheet provided by the test applicant.

^b 86% of maximum PTO power.

^c Computed as per Table 8 of PNS/BAFS PABES 301:2020, requirement if drawbar power is equal or below 65 kW.

^d Hydraulic lift test was not conducted as decided by the test applicant.

Note: ND – No Data

OBSERVATIONS

1. Operator's manual	Not provided
2. Previous test reports of similar brand and model	2020 – 0209
3. Manufacturer	FIRST TRACTOR COMPANY LIMITED 154 Jianshe Road, Luoyang, Henan, China
4. Serial number	42403738
5. Failures or abnormalities that may be observed on the machine or its components during and after the operation.	No failure or abnormalities were observed on the tractor or its component parts during and after the operation.
6. All tractors shall be equipped with Roll-Over Protective Structures (ROPS) and seat belts.	The four-wheel tractor is provided with ROPS and a seat belt.
7. Seat shall be provided which will adequately support the operator in all working and operating conditions. Adequate and comfortable support and protection for the feet shall also be provided. Seats and ROPS shall conform to the requirements specified in PAES 139 and 140: 2004.	a) The tractor seat is adjustable and provided with a seat belt. b) Adequate and comfortable support and protection for the feet are provided. c) The ROPS is free from manufacturing defects that may be injurious to the tractor operator, free from rust, and painted properly. The ROPS has two posts with a height and width of 860 mm and 70 mm, respectively. The canopy is bolted to the posts.
8. When the PTO is in use, a cover or casing that protects the sides of the PTO shaft shall be fitted. An additional non-rotating casing shall also be provided when the PTO is not in use. This casing shall enclose the PTO shaft completely and be fixed to the tractor body.	A PTO cover is provided to protect the sides of the PTO shaft when in use. A plastic PTO casing that can enclose the PTO shaft when not in use and can be fixed to the tractor is provided.
9. The engine of the tractor shall be equipped with cooling system suitable for tropical operations.	The engine of the tractor is provided with a water-cooling system.
10. Mechanism that minimizes/reduce vibration shall be provided.	It was not verified if the machine's vibration was minimized/reduced due to limited technical capabilities in quantifying and comparing vibration levels during the test operation.

OBSERVATIONS (Cont'n)

11. The four-wheel tractor shall be free from manufacturing defects that may be detrimental to its operation	The four-wheel tractor is free from any manufacturing defects.
12. Any uncoated metallic surface shall be free from rust and shall be painted properly	Any uncoated metallic surface of the tractor is properly painted and is free from rust.
13. Appropriate labels, safety symbols, and warnings shall be provided appropriately.	Provided
14. All hot surfaces that may come in contact with the operator shall be provided with heat shield or protective cover.	Provided
15. Attachment of ballast or other weights shall be provided, as necessary.	Provided
16. The four-wheel tractor shall be free from sharp edges and surfaces that may injure the operator. Warning notices shall be provided in accordance with PAES 330:2022.	a. The four-wheel tractor is free from sharp edges and surfaces that may injure the operator. b. Warning notices are affixed to the various parts of the tractor in the form of stickers. The warning notices are visible to the operator and indicate the particular danger that the operator is at risk of (Figure 6).
17. Other remarks	a) The hydraulic lift test was not conducted as decided by the test applicant. b) The engine was not disassembled as decided by the test applicant. c) The test applicant submitted an emission test report from China National Accreditation Service for Conformity Assessment (CNAS) Testing (Issued Date: 2016-7-15, Tested Model: YD4JZ55C3, Inspection Basis GB20891-2014 "Emission Limits and Measurement Methods for Pollutants from Non-road Mobile Machinery Diesel Engines (China Stage III and IV)")

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing
Dates of Test	July 30 – 31 & August 1, 2024
Location of Test	Roberto C. Bautista Hall, UPLB, College, Los Baños, Laguna
Items Tested	PTO Load Test, Traveling Speed, and Radius of Turning Circle/Area

METHODS OF TEST

The four-wheel tractor was tested based on the PNS/BAFS PABES 302:2020 – Agricultural Machinery – Four-wheel Tractor – Methods of Test.

DESCRIPTION OF THE MACHINE

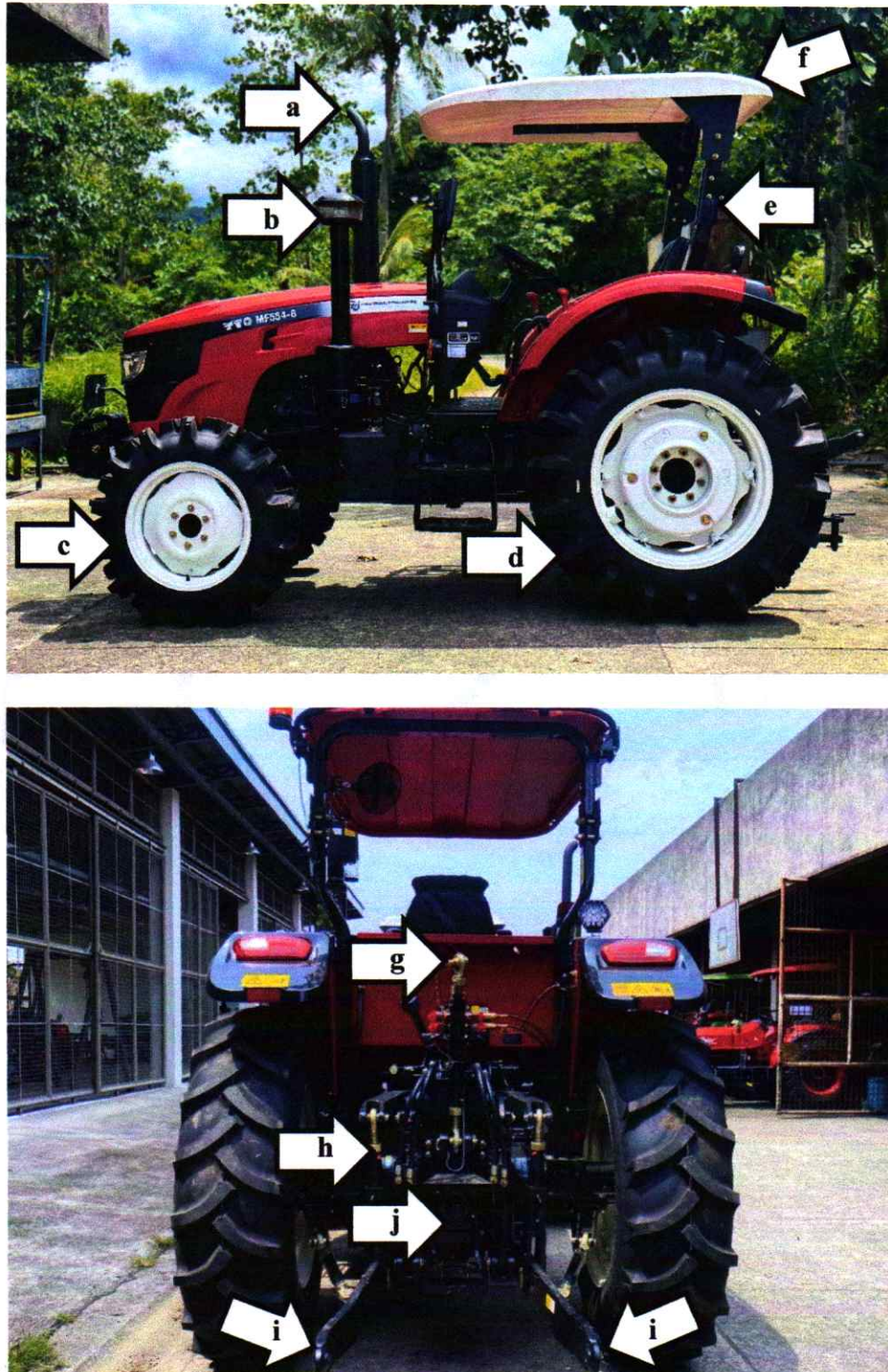


Figure 1. The different parts of the YTO MF554-6 TIER 3 Four-wheel Tractor include the following: (a) Exhaust, (b) Air cleaner with precleaner, (c) Front wheel, (d) Rear wheel, (e) ROPS, (f) Canopy, (g) Upper link, (h) Lift rod, (i) Lower links, and (j) PTO with cover and casing.

DESCRIPTION OF THE MACHINE (Cont'n)

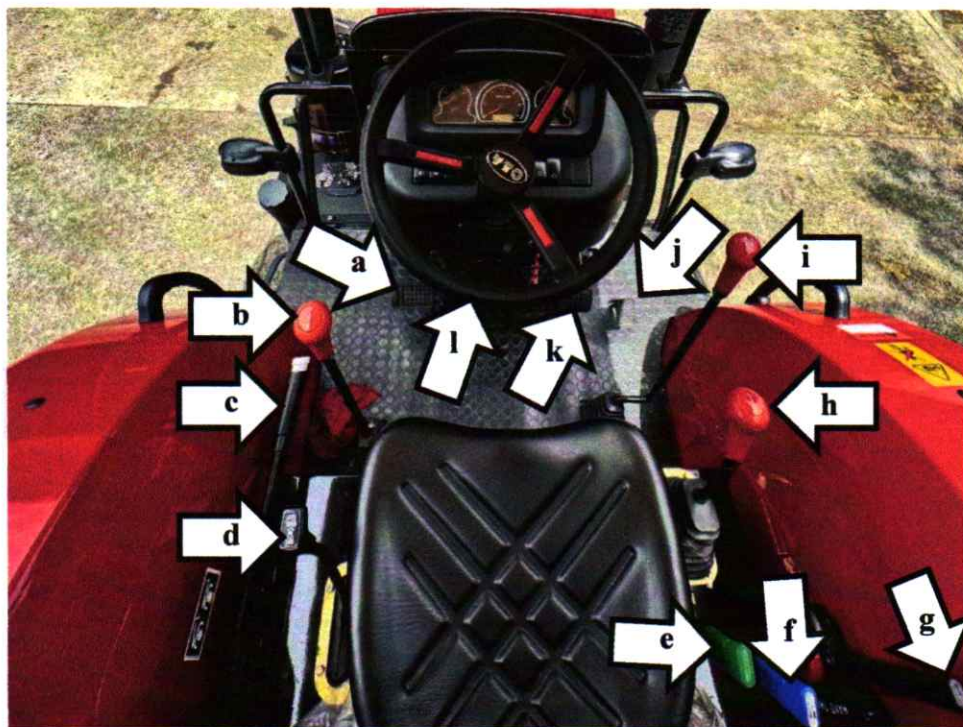


Figure 2. Controls of the YTO MF554-6 TIER 3 Four-wheel Tractor include the following:

- (a) Clutch pedal, (b) Shuttle gear lever, (c) PTO clutch lever, (d) PTO speed lever,
- (e) Draft control lever, (f) Position control lever, (g) Hand throttle,
- (h) Range gear lever, (i) Main gear lever, (j) Foot throttle,
- (k) Brake pedals, and (l) Steering wheel.

SPECIFICATIONS

Item	Manufacturer's Specification ^a	AMTEC Verification
1 Overall dimensions and weight of tractor		
1.1 Length, mm	3860	3775
1.2 Width, mm	1740	1652
1.3 Height, mm	2565	2489
1.4 Dry weight of tractor, kg	ND	NM ^b
2 Engine		
2.1 Brand	YTO	YTO
2.2 Make	China	China
2.3 Model	YD4JZ55C3	YD4JZ55C3
2.4 Serial number	Y221106792	Y221106792
2.5 Output power, kW	40.5	40.5 ^a
2.6 Rated shaft speed, rpm	2200	2200 ^a
2.7 Number of cylinders	ND	4
2.8 Bore × stroke, mm	ND	NM ^c
2.9 Displacement, cm ³	2980	NM ^c
2.10 Type	In-line, vertical, water-cooled, 4-stroke, Turbo charge	Four-stroke cycle, 4-cylinder, water-cooled, naturally-aspirated
2.11 Fuel used	ND	Diesel
2.12 Governor	ND	ND ^c
2.13 Air cleaner	ND	Oil-bath type
2.14 Lubrication system	ND	ND ^c
2.15 Cooling system	Water-cooled	Water-cooled
2.16 Starting system	ND	Electric start
2.17 Electrical system	ND	Battery-operated
2.18 Exhaust system	ND	Vertical muffler
3 Steering system	ND	Full hydrostatic

^a The data were obtained from the tractor engine nameplate and the specifications sheet provided by the test applicant.

^b There is no available instrument to facilitate the measurement of the parameter during the test operation.

^c The engine was not disassembled as decided by the test applicant.

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
4	Power take-off		
4.1	Location	Rear-mounted	Rear of the tractor
4.2	Type	Independent	Type I
4.2.1	Diameter of PTO shaft ends, mm	ND	34.63
4.2.2	Number of splines	ND	6
4.2.3	Rated shaft speed, rpm	540/1000	540
4.3	Height above ground, mm	ND	605
4.4	Direction of rotation (viewed from the rear of the tractor)	ND	Clockwise
4.5	Mode of operation		
4.5.1	Speed change	ND	Lever-operated
4.5.2	Clutch	ND	Independent
4.6	Rated PTO Power, kW	38.5	42.49
5	Ground clearance, mm	377	143
6	Three-point linkage		
6.1	Category		
6.1.1	Based on PTO power at rated rotational frequency of the engine	ND	1, 2N/2
6.1.2	Based on dimensions of the tractor linkage point	ND	2N/2
6.2	Length of lift arms, mm	ND	224
6.3	Length of lower links, mm	ND	766
6.4	Horizontal distance between the two lower links, mm	ND	865
6.5	Horizontal distance between the two lift arms endpoints, mm	ND	493
6.6	Length of upper link, mm		
6.6.1	Minimum	ND	510
6.6.2	Maximum	ND	762
6.7	Distance of lower link pivot point to lift rod pivot points on lower links, mm	ND	350

^a The data were obtained from the specifications sheet provided by the test applicant.

Note: ND – No Data

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
6.8	Length of lift rods, mm		
6.8.1	Minimum	ND	554
6.8.2	Maximum	ND	715
6.9	Diameter of top link pin hole, mm	ND	25.36
6.10	Diameter of lower link pin hole, mm	ND	24.98
7	Drawbar		
7.1	Type	ND	Fixed
7.2	Category based on PTO power	ND	Category 1, 2, 3, 4, and 5
7.3	Drawbar width, mm	ND	71.63 (Upper) 71.05 (Lower)
7.4	Drawbar thickness, mm		
7.4.1	Upper clevis	ND	22.53
7.4.2	Lower clevis	ND	30.10
7.5	Pin hole diameter, mm	ND	36.13 (Upper) 35.63 (Lower)
7.6	Pin diameter, mm	ND	34.10
7.7	F ^b , mm	ND	37.14
7.8	G ^b , mm	ND	403
7.9	Height, mm	ND	67.08
7.10	Throat depth, mm	ND	91.41
7.11	End radius of drawbar and clevis, mm	ND	NM
7.12	W ^b , mm	NA	NA

^a The test applicant did not submit complete machine specifications.

^b Based on Table 6 of PNS/BAFS PABES 301:2020.

Note: NA – Not Applicable (The ends of the drawbar are not chamfered), ND – No Data

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
8	Transmission system		
8.1	Main clutch and PTO clutch	ND	Independent
8.2	Transmission gears		
8.2.1	Type	ND	Synchromesh
8.2.2	Main gear	ND	1, 2, 3, 4, N
8.2.3	Range gear	ND	Low, High
8.2.4	Shuttle gear	ND	Forward, Reverse
8.3	Differential lock	ND	Pedal-type
9	Tire		
9.1	Type	ND	Bias-ply
9.2	Front tire size, W × D, mm	8.3-20	211 × 890
9.3	Rear tire size, W × D, mm	12.4/11-28	315 × 1199
9.4	Wheelbase, mm	ND	2105
9.5	Wheel tread		
9.5.1	Front tread, mm	ND	1312
9.5.2	Rear tread, mm	ND	1315
10	Brake system		
10.1	Type according to manner of applying braking force	ND	Hydraulic
10.2	Type according to manner of transmitting the force from the controls	ND	Mechanical
10.3	Parking brake	ND	Lever-operated

^a The data were obtained from the component markings. The test applicant did not submit complete machine specifications.

Note: ND – No Data

RESULTS

Table 1. Radius of turning circle/area of the YTO MF554-6 TIER 3 Four-wheel Tractor.

Item	Without Brake		With Brake	
	Right Turn	Left Turn	Right Turn	Left Turn
Radius of Turning Circle, mm	4273	4183	3928	3930
Radius of Turning Area, mm	4378	4288	4033	4036

Table 2. Traveling speed of the YTO MF554-6 TIER 3 Four-wheel Tractor at different gear settings.

Range Gear Settings	Main Gear Settings	Shuttle Gear Settings	
		Forward (kph)	Reverse (kph)
Low	First	2.69	2.48
	Second	3.59	3.21
	Third	5.45	5.01
	Fourth	7.59	7.11
High	First	11.85	10.55
	Second	15.62	13.90
	Third	24.62	22.49
	Fourth	29.62	27.16

RESULTS (Cont'n)

Table 3. Test conditions during the power take-off load test of the YTO MF554-6 TIER 3 Four-wheel Tractor.

Temperature, °C	32.5
Relative Humidity, %	61.6

Table 4. Power Take-Off Load Test data of the YTO MF554-6 TIER 3 Four-wheel Tractor.

Speed		PTO		Fuel Consumption, L/h	Specific Fuel Consumption, g/kW-h
Engine, rpm	PTO, rpm	Torque, N-m	Power, kW		
2325	665	47	3.27	5.09	1307.74
2308	660	235	16.20	7.43	385.23
2271	640	513	34.38	11.10	271.17
2245	619	605	39.19	11.81	253.11
2090	600	651	40.88	11.91	244.72
2048	580	688	41.78	11.98	240.84
2004	560	725	42.49	11.98	236.83
1860	543	741	42.07	11.75	234.63
1801	520	753	41.01	11.30	231.48
1704	481	781	39.33	10.68	228.12

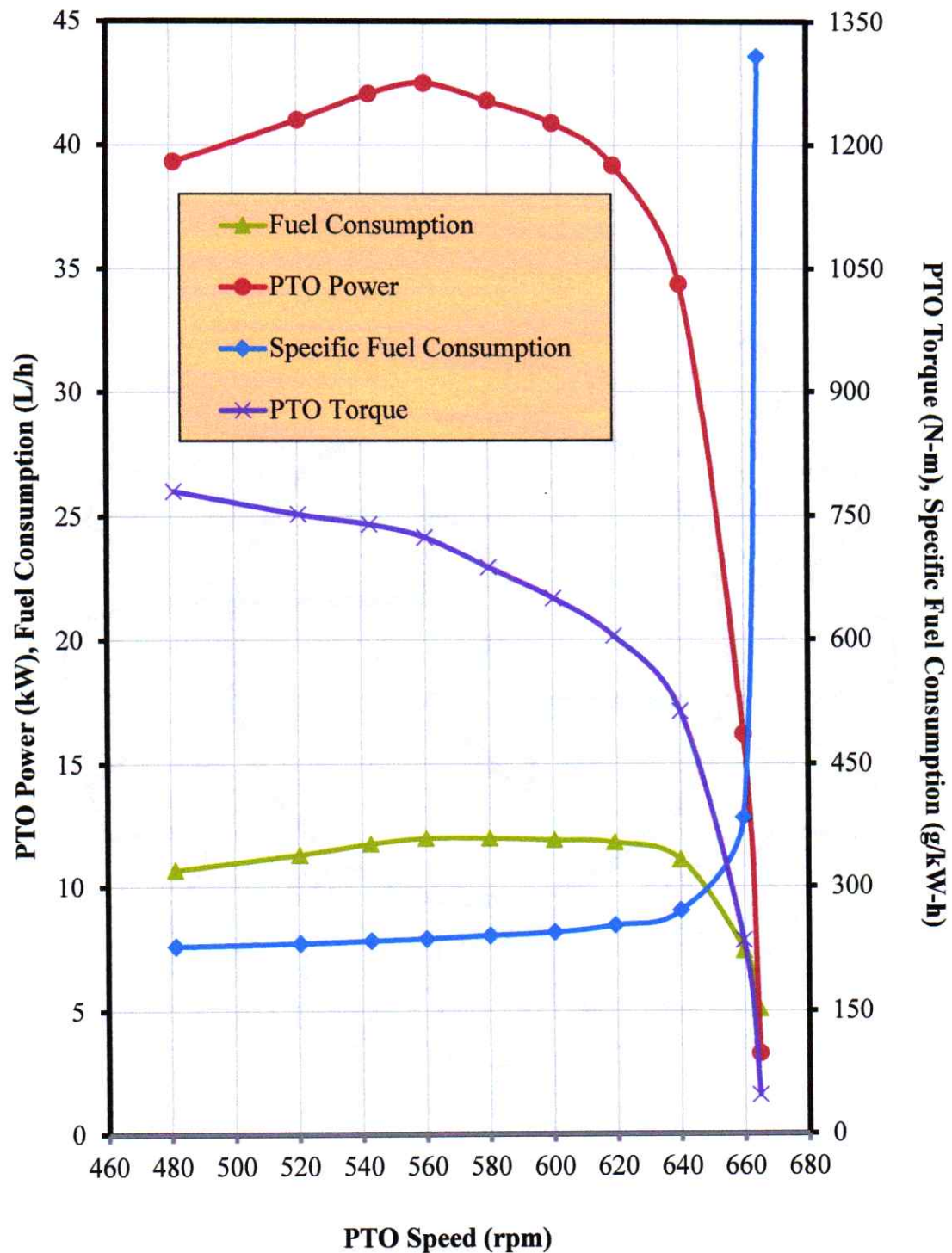
RESULTS (Cont'n)

Figure 3. Performance characteristics curves of the YTO MF554-6 TIER 3 Four-wheel Tractor.

OBSERVATIONS



Figure 4. Nameplate of the YTO MF554-6 TIER 3 Four-wheel Tractor.



Figure 5. Nameplate of the engine of the YTO MF554-6 TIER 3 Four-wheel Tractor.

OBSERVATIONS (Cont'n)

Figure 6. Markings, labeling, and safety warning stickers on the YTO MF554-6 TIER 3 Four-wheel Tractor.

Tested by:

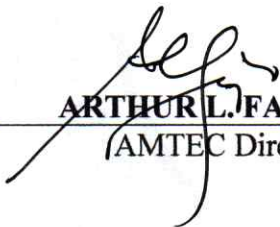

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29 JUL 2025
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